



University of Engineering and Technology
Mardan

HCI Lab_02

Ali Khan – 23MDSWE248

SE-401L Human Computer Interaction Lab

Sir Dr. Hamid Ullah

19 FEB 2026



Task#1: Figure 1.2 is a screenshot of a desktop application in which you can select a certificate template and then print it. To select a template, you go through the scroll bar and then choose the template that you like. Discuss if this is a good design, if not what are some of the main problems with the design. Think about a situation where you like a design and print one certificate and then later you come back to print the same certificate. How easy or difficult would it be to re select the same design that you have already chosen before.

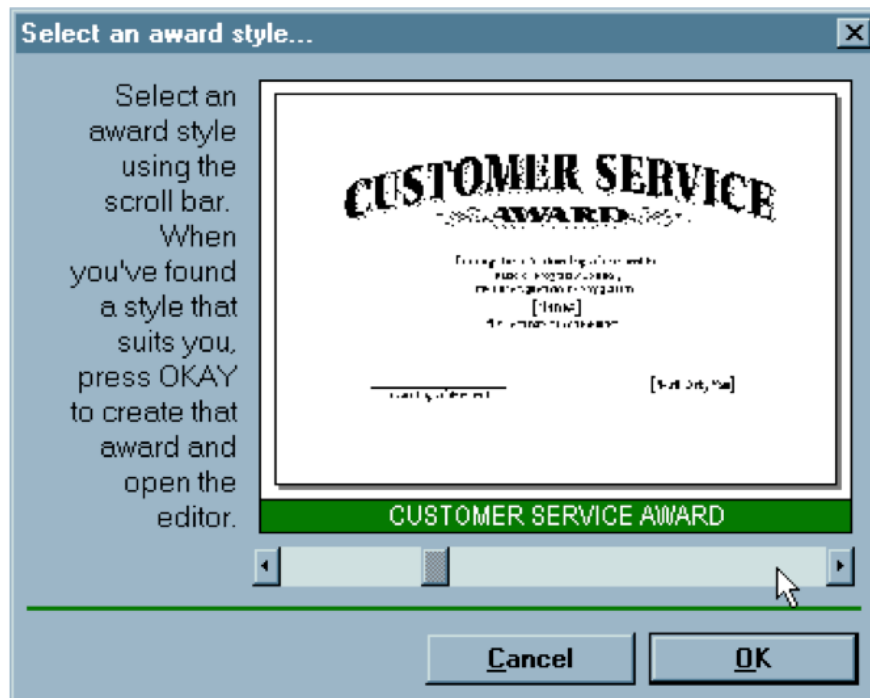


Figure 1.2: Desktop Application for Certificate Printing

Solution:

Design Aspect	Current State (Problem)	HCI Principle Violated	Impact on User Experience
Navigation Model	Linear/Sequential (one-by-one scrolling).	Efficiency of Use	High interaction cost; users must pass every item to reach the end.

Visual Scanning	Only one template visible at a time.	Recognition vs. Recall	Impossible to compare styles side-by-side; forces mental effort to remember previous options.
Searchability	No search bar, categories, or filters.	Flexibility & Efficiency	Users cannot jump directly to a "Birthday" or "Academic" style.
System Feedback	Scroll bar lacks "Page X of Y" or count indicators.	Visibility of System Status	User has no idea how many templates exist or how much progress they've made.
Re-selection Task	No "Recently Used" or "Favorites" section.	User Memory Load	Returning users must scroll and "eye-ball" it to find their previous choice, leading to errors.
Control	Horizontal scroll bar for high-detail thumbnails.	Fitts's Law	Small scroll arrows are "hard to hit" targets; dragging the slider for 100+ items is imprecise.

Task#02: The following screenshots (Figure 1.3 to 1.6) of Dominos website are related to placing order followed by the checkout. Please note that a registered customer (of the website) is performing this task.

Solution:

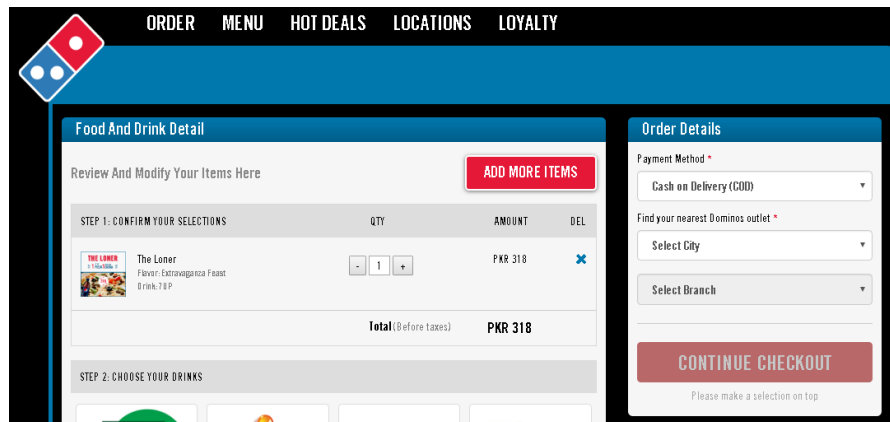


Figure 1.3 Check Out Page

Feature	Analysis	HCI Evaluation
Branch Selection	The system requires manual selection of City and Branch even for a registered user.	High Interaction Cost: Forces a logged-in user to perform "Search" tasks that should be automated by their profile.
Checkout Button	The "Continue Checkout" button is grayed out and contains the text "Please make a selection on top".	Visibility of System Status: Provides clear feedback on why the user cannot proceed yet.
Color of Checkout Button	The check out button color is red	Red color conveys a danger message the color of the button should be green

The screenshot displays the Domino's website checkout process. At the top, a navigation bar includes links for ORDER, MENU, HOT DEALS, LOCATIONS, and LOYALTY. The main content is divided into two columns. The left column, titled 'Food And Drink Detail', shows a review of items under 'STEP 1: CONFIRM YOUR SELECTIONS'. It lists 'The Loner' (Flavor: Extravaganza Feast, Drink: 7UP) with a quantity of 1 and a price of PKR 318. Below this, 'STEP 2: CHOOSE YOUR DRINKS' shows four drink options: 7UP, Fanta, Sprite, and Mountain Dew. The right column, titled 'Order Details', shows the payment method as 'Cash on Delivery (COD)', the location as 'Islamabad', and the branch as 'Domino's G11 Branch'. It also includes a 'Select Area' dropdown menu with 'G11 Markaz' selected. A 'CONTINUE CHECKOUT' button is at the bottom right.

Figure 1.4 City and branch selection

Feature	Analysis	HCI Evaluation
Dropdown Logic	Once the Branch is selected, a new "Select Area" dropdown appears.	Step-wise Disclosure: While it keeps the UI clean, it adds unexpected steps to the user's mental model of the checkout flow. If there is only one area auto set
Consistency	The layout shifts slightly as new fields (G11 Markaz) are revealed.	Flexibility: This rigid hierarchy prevents users from quickly changing a city without resetting the entire branch/area chain.

The screenshot shows a web interface with a dark blue header. On the left, there's a 'YOUR INFORMATION' section with fields for First Name (Absar), Last Name (Shahid), Email Address (redacted), and Callback Phone # (placeholder: Enter Callback Phone #). Below this is a 'SHIPPING INFORMATION' section with City/Town (Islamabad) and Address (redacted). A red error message 'Please select the city from suggested list to proceed.' is visible under the City/Town field. On the right, there's a summary table with Subtotal (PKR 318), Taxes (PKR 51), Shipping (Free), and Order total (PKR 369). At the bottom right is a large red 'PLACE ORDER' button.

YOUR INFORMATION	
* First Name	Absar
* Last Name	Shahid
* Email Address	[Redacted]
* Callback Phone #	Enter Callback Phone #

SHIPPING INFORMATION	
* City/Town	Islamabad
Please select the city from suggested list to proceed.	
* Address	[Redacted]

Subtotal	PKR 318
Taxes	PKR 51
Shipping	Free
Order total	PKR 369

PLACE ORDER

Figure 1.5 Contact information option

Feature	Analysis	HCI Evaluation
Validation Error	Red text appears under "City/Town" despite "Islamabad" being visible in the field.	System Inconsistency: The system displays a value but simultaneously treats it as empty or invalid, causing user confusion.
Missing Profile Data	The "Callback Phone #" field is empty even though the user is registered and their name/email are present.	User Memory Load: Fails to "Respect User Data"; forcing a registered user to re-enter a primary contact number is a major friction point.
Visual Hierarchy	The "Place Order" button is dark and inactive.	Error Prevention: Prevents the user from submitting a broken form, but the reason for the "City" error is not visually intuitive.

YOUR INFORMATION

* First Name

* Last Name

* Email Address

* Callback Phone #

SHIPPING INFORMATION

* City/Town

* Address

Promotional Code

Taxes PKR 51

Shipping Free

Order total PKR 369

PLACE ORDER

Figure 1.6 Shipping information

Feature	Analysis	HCI Evaluation
Hidden List	The user must click the already-filled "Islamabad" box to reveal a list of geographic suggestions.	Recognition vs. Recall: The system forces the user to "recognize" the city from a global list (including India and Bangladesh) rather than accepting the valid local text.
Ambiguity	Suggestions include "Islamabad, Punjab, Pakistan" and "Islamabad, Uttar Pradesh, India".	Information Scent: For a local pizza delivery, providing international city options increases the risk of "Slip" errors (accidentally clicking the wrong Islamabad).

Task 03: You are looking for a bargain to buy some clothes for your niece who is five years old. You are on a budget so you are looking in the sales section.

- Visit the following website:

<https://www.minnieminors.com/pages/minnie-minors>

You are only allowed to explore the sales section of the website. Go through the process except for checkout and list down the problems that you encountered.

- Now visit the following and repeat the same process.

<https://www.ilovehopscotch.com/new-in.html>

Which one is better in your opinion in terms of ease of finding what you are looking for (Not the quality of the merchandise)

Solution:

1. MinnieMinors:

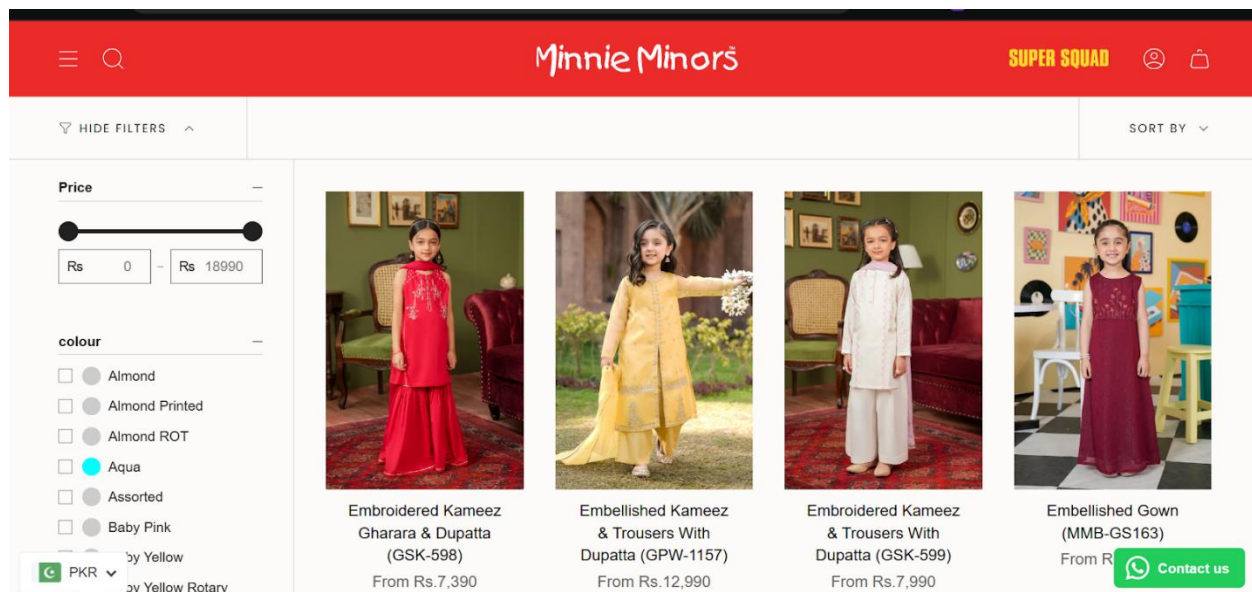


Figure 3.1

Feature	Analysis	HCI Evaluation	Fig
Sales Page button	Not Visible	As we are on budget looking for product on sales so we will not buy from here because it does not have a sales page button or we cannot see which products are on sales	3.1

2. HopsCotch:

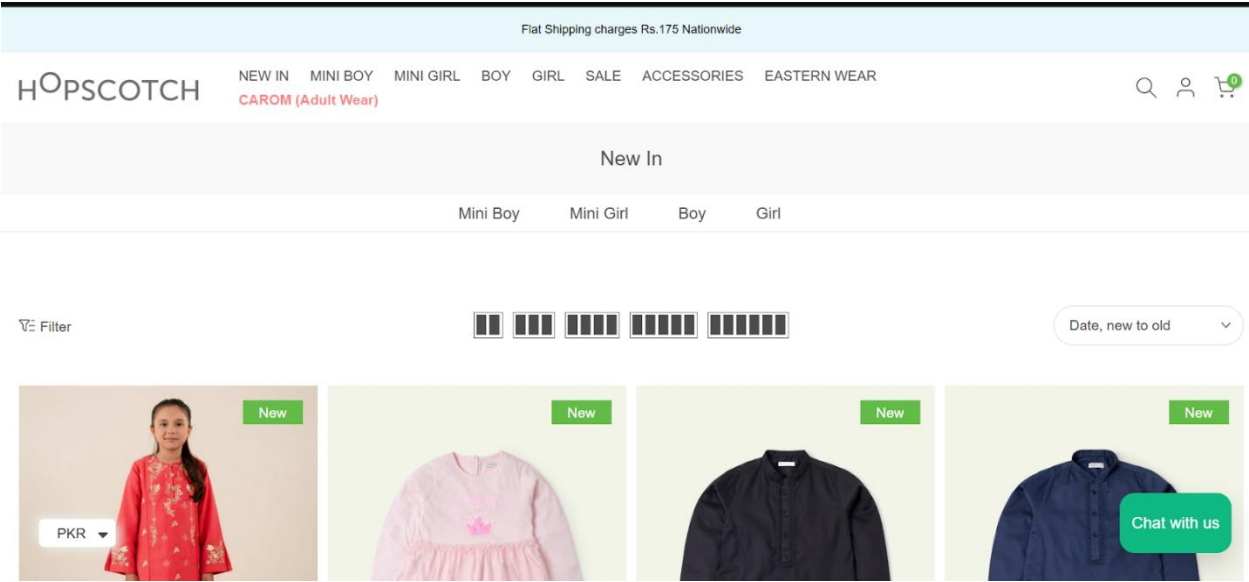


Figure 3.2.1

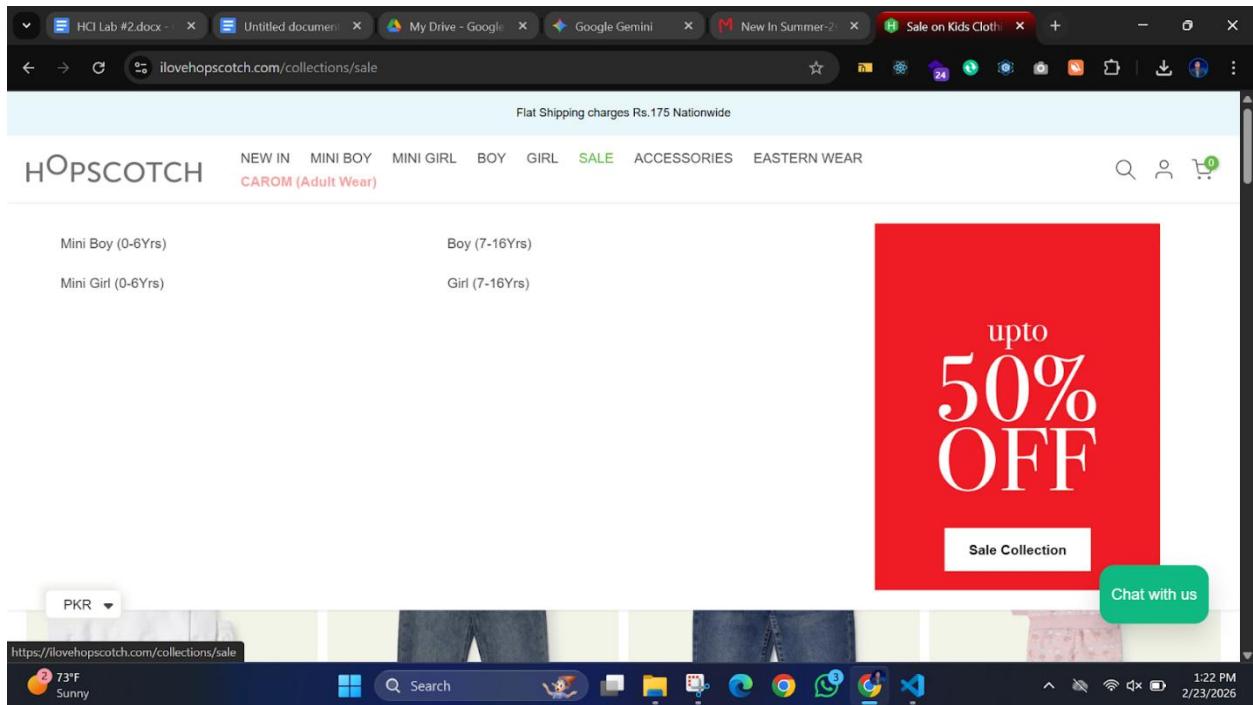


Figure 3.2.2

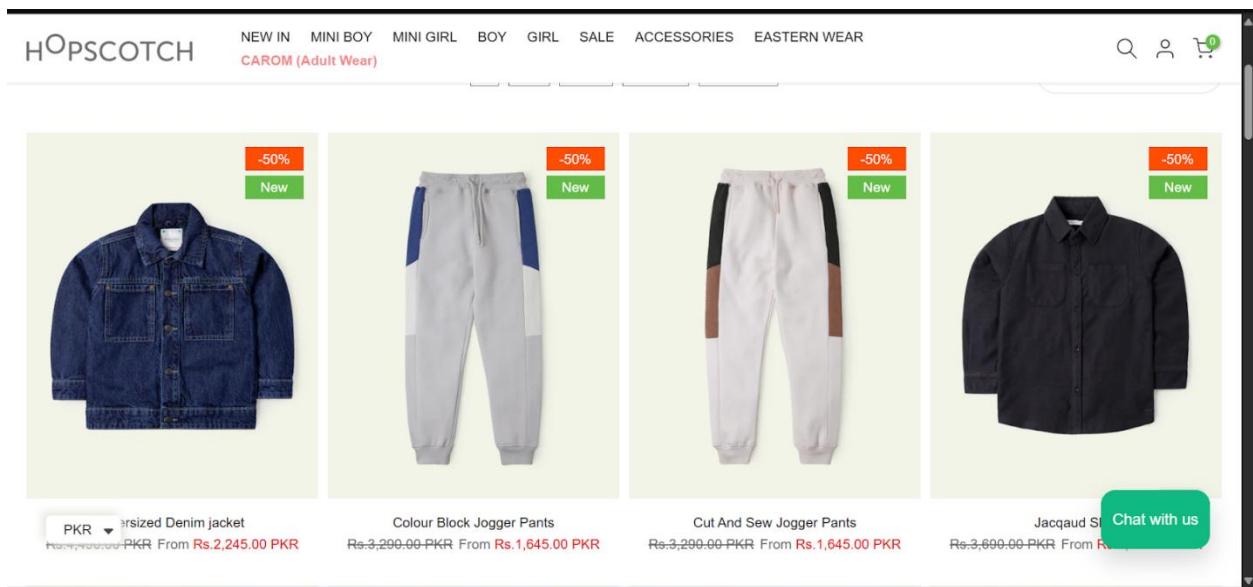


Figure 3.2.3

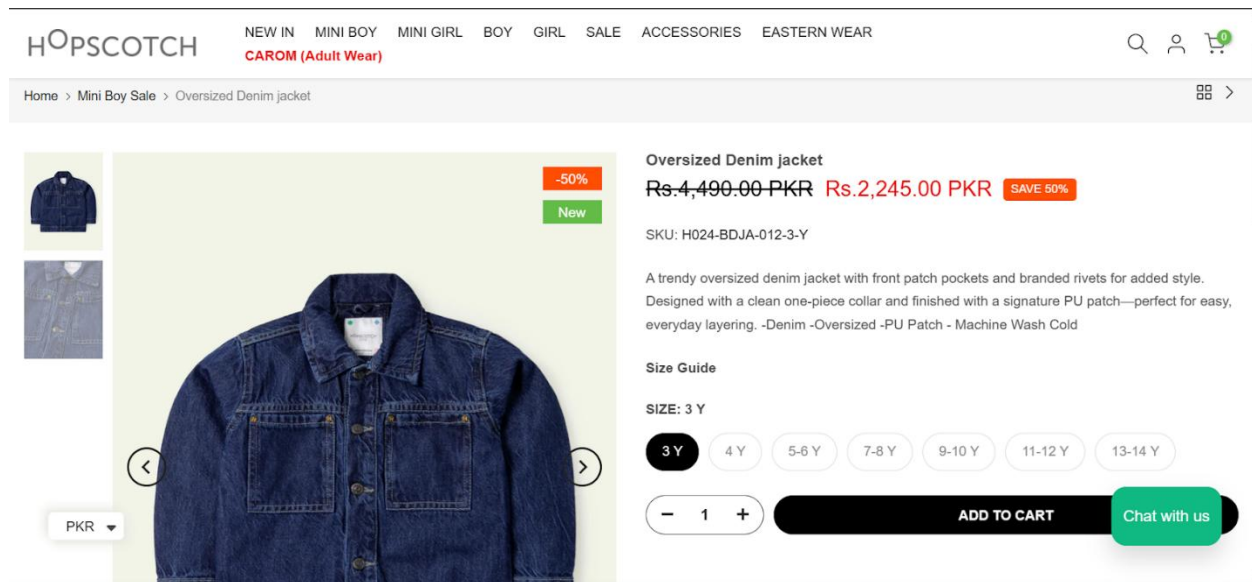


Figure 3.2.4

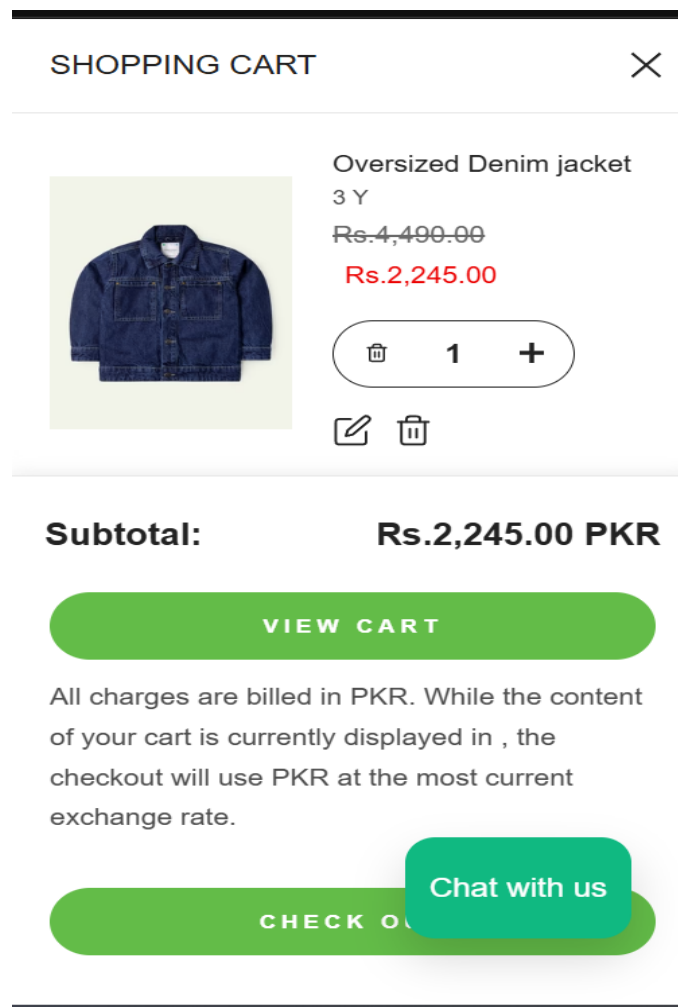


Figure 3.2.5

Feature	Analysis	HCI Evaluation	Fig
Sales button	Visible	Clearly show sales page so we will like to buy from here	3.2.1
Hover on Sale	Shows a drop down	Clearly shows a drop down where we can see different ages range on which there is sale	3.2.2
ADD to cart page	Shows product details and add to cart button	Product image is zoomable which is good but add to cart button is half hidden by WhatsApp sticky icon	3.2.3
Check out bar	Shows product price and check out button	Check out button is hidden by WhatsApp icon	3.2.4

Task 04: Compare two university websites. Take any website from Pakistan and one from USA. Its, easier to access a system if we have a specific task in mind. So, imagine you are going for your masters and you need to make up your mind which degree to get, which website makes it easier to get the information and also which one is simpler in understanding the process of admission. Please include screenshots.

Solution:

Activity: Search for Admission in MS Computer Software Engineering or Computer Science

1. UET Mardan

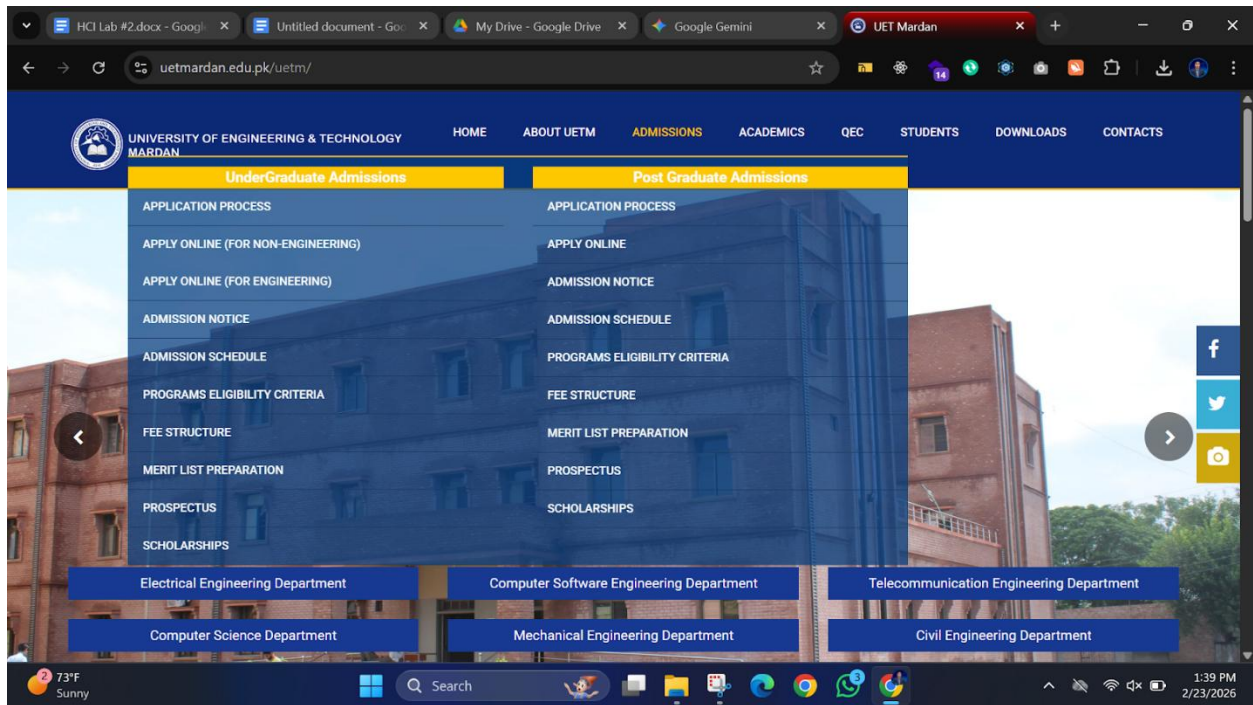


Figure 4.1.1

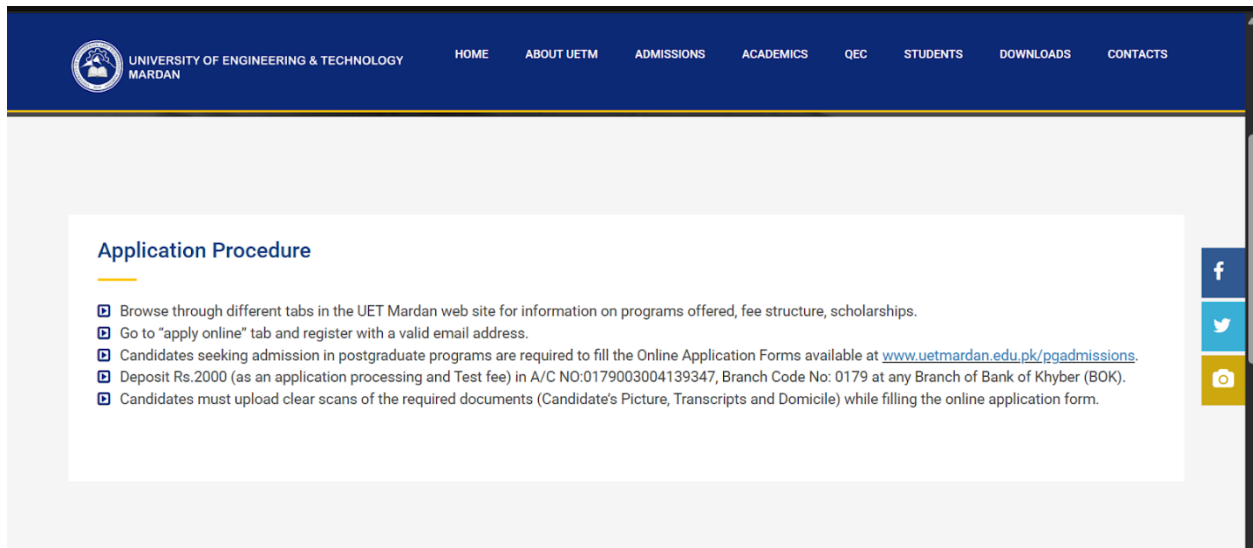


Figure 4.1.2

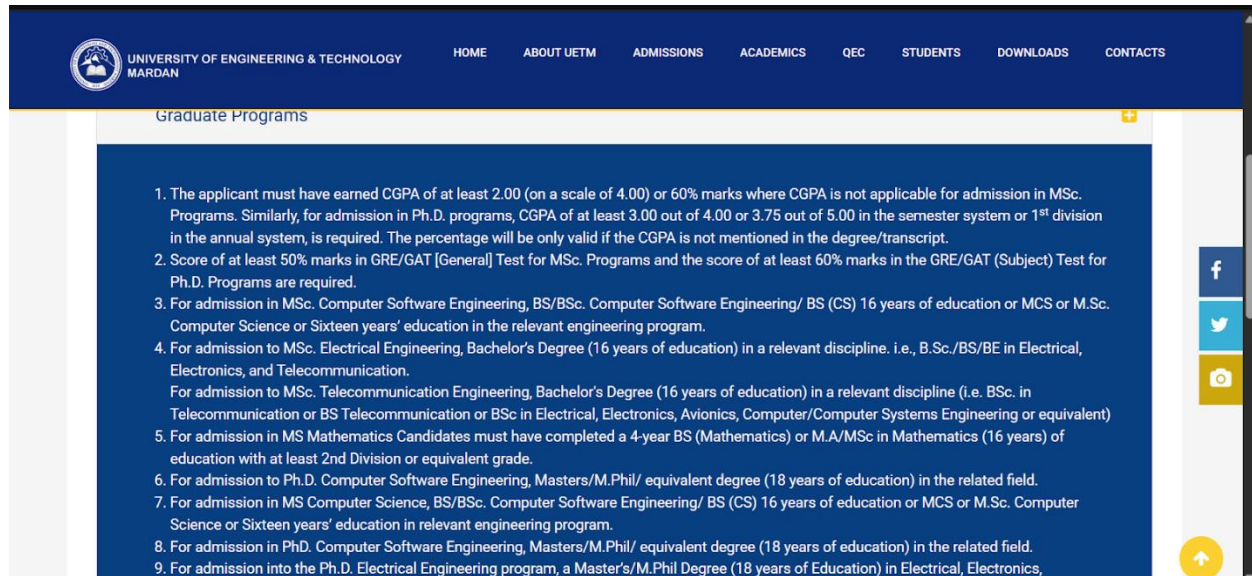


Figure 4.1.3

Feature	Observations	HCI Analysis	Fig
Finding Programs	Programs are listed under "Departments." You have to click a specific department (e.g., Computer Software Engineering) to see if they offer an MSc.	High Interaction Cost: There is no central "Graduate Programs" directory. Users must hunt through departments to find degree options.	4.1.1
Admission Process	Detailed in a "News" or "Notice" format with a specific timeline (e.g., "MSc/MS/PhD Admissions Spring 2026").	Time-Sensitivity: The site relies on "Latest News" for process info. If you visit in the off-season, finding the <i>general</i> process is difficult.	4.1.2

Instructions	High detail on bank deposit accounts, manual document scans, and specific dates.	Cognitive Load: The instructions are text-heavy and look like a legal document. This requires high concentration to ensure no steps are missed.	4.1.3
Mental Model	Follows a "Notice Board" model.	Traditional: Great for local students used to the system, but difficult for an outsider to navigate quickly.	
Fields of degree	Not present	It does not contain a categories/fields of MS studies No easy avigation	4.1.1, 4.1.2, 4.1.3

2. Arizona State University

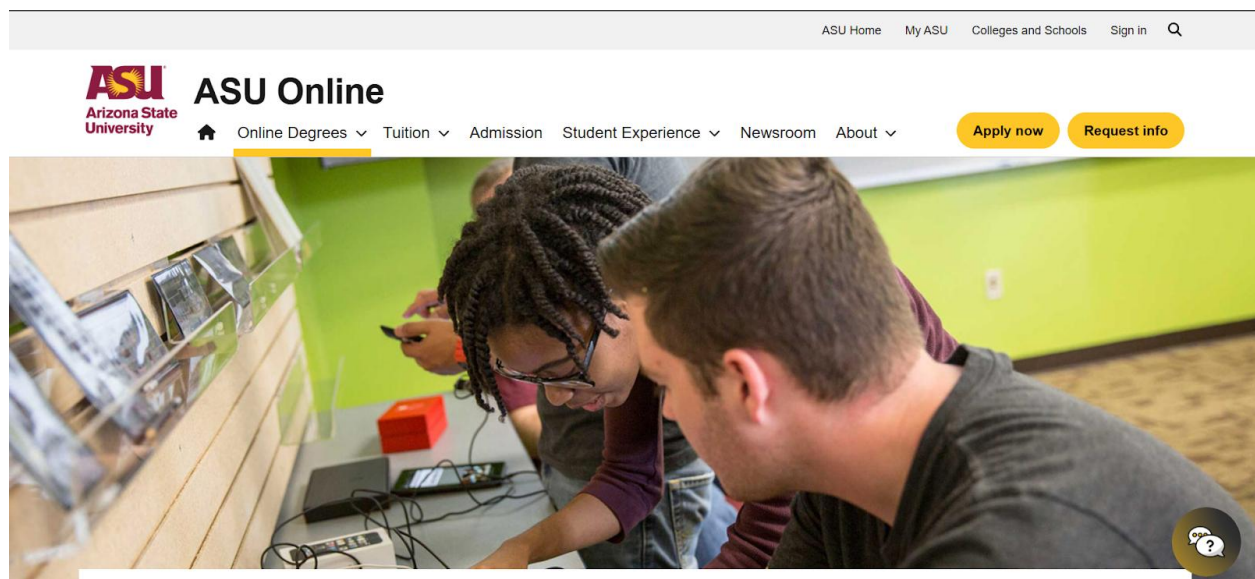


Figure 4.2.1

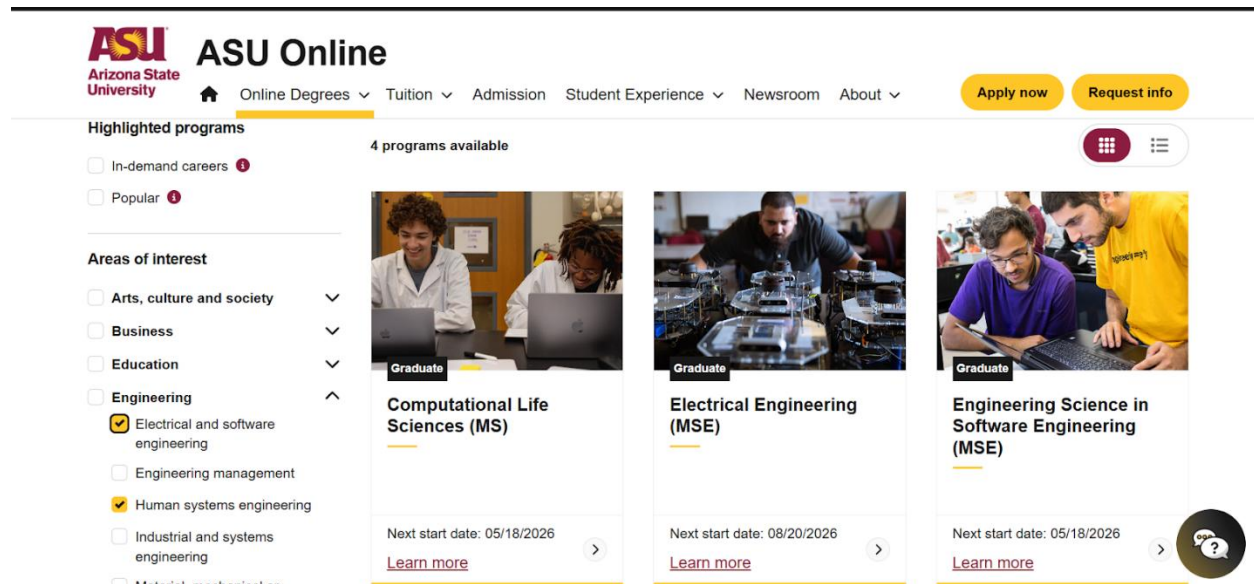


Figure 4.2.2

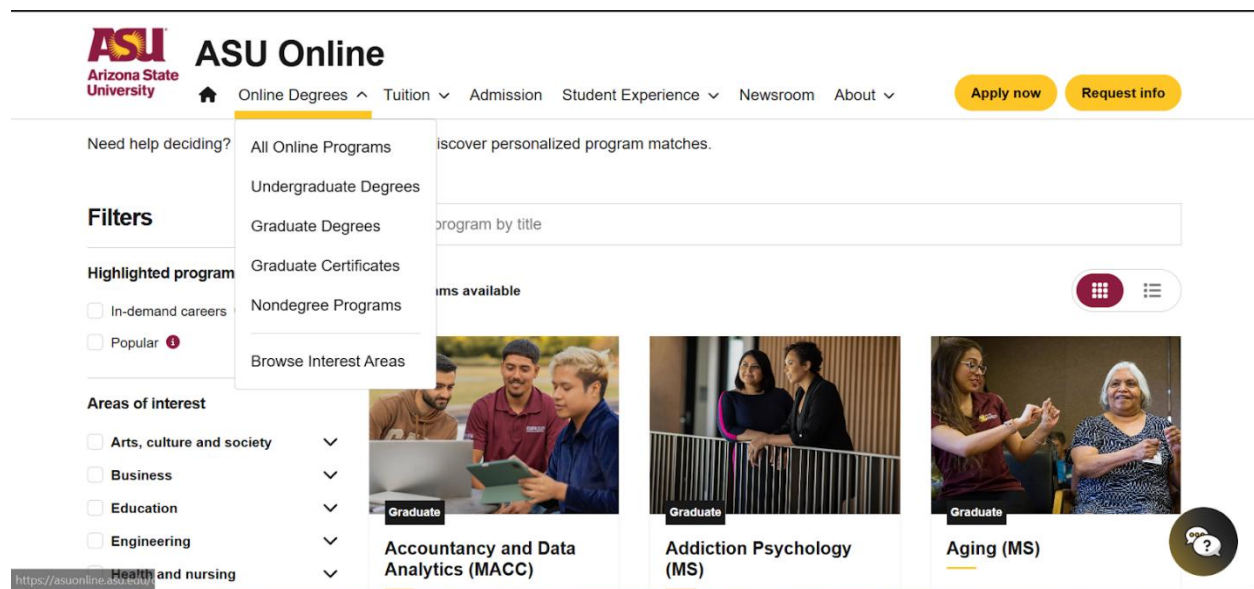


Figure 4.2.3

Feature	Observations	HCI Analysis	Fig
Finding Programs	Features a "Degree Finder" or "All Graduate Programs" list with filters for "Interest	Information Scent: Users can find their degree in seconds using keywords.	4.1

	Area" (e.g., Engineering, UX).	This matches the user's intent immediately.	
Admission Process	Broken down into 3–4 visual steps: "Requirements," "Application," "Transcripts".	Chunking: By breaking complex tasks into small, digestible chunks, the site reduces user anxiety and makes the process feel "doable."	4.2
CTAs (Call to Action)	Large "Apply Now" or "Request Info" buttons follow the user as they scroll.	Affordance: The system clearly signals the "next step" at all times, preventing the user from getting lost.	4.3
Mental Model	Follows a "Product/Service" model.	User-Centric: The website acts as a guide, actively helping the user move through the funnel.	
Field of MS/ Area of MS	Categories dropdown	Clearly shows categories drop down for MS admissions in different fields	

THE END